

# WRITE & SOLVE LINEAR SYSTEMS IN CONTEXT

The image displays three sample cards for the 'Problem Pass' activity, each with a title 'Problem Pass Write & Solve Linear Systems In Context' and a problem statement. The cards are tilted and arranged in a cluster. The left card contains 'Problem 2' about Roy and Karen reading a book. The right card contains 'Problem 3' about Suzanne and Kaitlin's ages. The bottom card contains 'Problem 1' about Erica's Dessert Shop. Each card has a space for the student to write and solve the system of equations. A yellow starburst on the left says 'Algebra 1'. A purple box on the right says '8 Rounds'. The bottom of the cluster has the title 'Problem Pass Activity' and the author 'Made by Debbie Veatch'.

**Problem Pass**  
**Write & Solve Linear**  
**Systems In Context**

**Problem 2:**  
Roy and Karen are both reading a book for their English Literature class. Roy has already read 110 pages and plans to read 45 pages a day. Karen has read 305 pages and plans to read 30 pages a day.

Write and solve a system to represent this situation. How many days will they have read the same number of pages? What will be the number of pages?

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**Problem Pass**  
**Write & Solve Linear**  
**Systems In Context**

**Problem 3:**  
Suzanne's age is two less than six times Kaitlin's age. Together their ages add up to 82.

Write and solve a system of equations to determine each person's age.

Algebra Activities

**Problem Pass**  
**Write & Solve Linear**  
**Systems In Context**

**Problem 1:**  
Erica's Dessert Shop sells a slice of cheesecake for \$7 and a slice of pie for \$4. Yesterday she sold 205 slices of dessert for a total of \$1180.

Write and solve a system of equations to determine the number of slices of each type of dessert she sold.

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**Algebra 1**

**8 Rounds**

**Problem Pass Activity**  
Made by Debbie Veatch

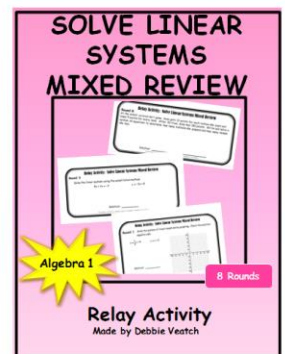
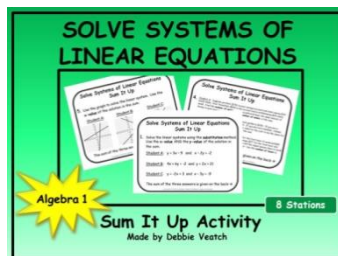
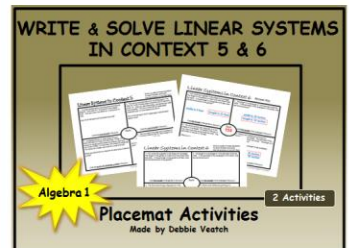
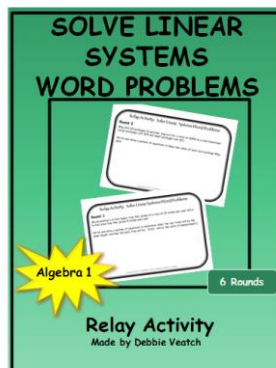
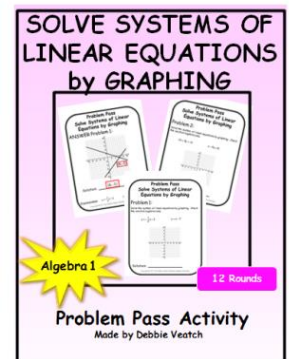
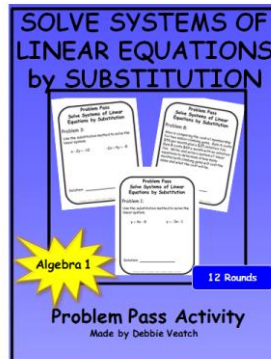
# Thank You!

Thank you for downloading my product. I have personally used activities I have created in my own classroom and can testify to their effectiveness. I truly appreciate the support and look forward to your feedback.

More activities are coming soon, so please check back often.

Debbie

## You might also like:



More activities coming soon . . .

Please check back often!

# Write & Solve Linear Systems In Context

## Problem Pass Activity

### Objective:

To practice or review writing and solving systems of linear equations word problems. The activity includes problems with mixtures, comparing two deals, finding the cost, age and geometry.

### **CCSS: Analyze and solve pairs of simultaneous linear equations.**

8.EE.C.8 Analyze and solve pairs of simultaneous linear equations.

8.EE.C.8a Understand that solutions to a system of two linear equations in two variables correspond to points of intersection of their graphs, because points of intersection satisfy both equations simultaneously.

8.EE.C.8b Solve systems of two linear equations in two variables algebraically, and estimate solutions by graphing the equations. Solve simple cases by inspection.

8.EE.C.8c Solve real-world and mathematical problems leading to two linear equations in two variables.

### **CCSS: Solve systems of equations.**

HSA.REI.C.5 Prove that, given a system of two equations in two variables, replacing one equation by the sum of that equation and a multiple of the other produces a system with the same solutions.

HSA.REI.C.6 Solve systems of linear equations exactly and approximately (e.g., with graphs), focusing on pairs of linear equations in two variables.

### **CCSS: Create equations that describe numbers or relationships.**

HSA.CED.A.3 Represent constraints by equations or inequalities, and by systems of equations and/or inequalities, and interpret solutions as viable or nonviable options in a modeling context.

### Activity Directions:

Students work with a partner while seated at their desks. They should write in their own notebook or on the blank Answer Sheet (included).

At the same time the first pair of students is working on Problem 1, the second pair works on Problem 2, the third pair works on Problem 3, etc. At the teacher's signal, all students pass their problem in a specific direction. The students who started with Problem 2 should now pass it to the students who started with Problem 1, the students who started with Problem 3 pass it to the students who started with Problem 2, etc. The first pair of students starting with Problem 1 should deliver their finished problem to the last pair of students, or the teacher may prefer to deliver these each time. Students now flip over their new page to find the ANSWER to the problem they just finished.

**Activity Directions continued →**

# Write & Solve Linear Systems In Context

## Problem Pass Activity

### Activity Description, cont:

Students continue to work problems in order, pass problems and check their answers on the back of the next problem page until they have completed all problems included in the activity.

It is important for the teacher to tell the entire class when to pass their first problem and to tell everyone where to look for the first answer on the back of their NEXT problem. After this first pass together, I've allowed students to work and pass problems at their own pace. Teachers may prefer to set a timer so all students pass their current problem to the next pair of students simultaneously.

### Teacher Setup:

Print single-sided copies and slide pages into plastic page protectors to keep problems and answers together. Put Problem 1 and ANSWER Problem 8 back-to-back in the same plastic page protector; put Problem 2 and ANSWER Problem 1 back-to-back in another plastic page protector, Problem 3 and ANSWER Problem 2 together, etc. Prepare 2 or more complete sets of the activity to have enough pages for each pair of students in the class. Keep complete sets in order.

**It is very important to hand out problems in numerical order so the page with the answer on the back follows its problem number.** As you hand out problems in order, problem side up, tell each pair of students the direction they should pass their problem when finished. This direction may vary by row if you zig-zag or "snake" up and down the rows of desks. The first pair of students starting with Problem 1 should deliver their finished problems to the last pair of students, or you may prefer to deliver these each time.

(Optional) Print enough copies of the blank Answer Sheet for each student to use as they work the problems. Another option is for students to write in their own notebooks.

### Included in the package:

- Eight "Problem Pass" pages
- Answer pages for students to self-check
- Blank answer sheet for students (optional)
- Answer Key for Teacher

**This purchase is for one teacher only.**

This resource is not to be shared with colleagues or used by an entire grade level, school, or district without purchasing the proper number of licenses. If you are a coach, principal, or district interested in a site license, please contact me for a quote at [debbiesalgebraactivities@gmail.com](mailto:debbiesalgebraactivities@gmail.com). This resource may not be uploaded to the internet in any form, including classroom/personal websites or network drives.

# Printing Instructions

- Print single-sided copies and slide pages into plastic page protectors to keep problems and answers together.
- Put Problem 1 and ANSWER Problem 8 back-to-back in the same plastic page protector; put Problem 2 and ANSWER Problem 1 back-to-back in another plastic page protector; Problem 3 and ANSWER Problem 2 together, etc.
- Prepare 2 or more complete sets of the activity to have enough pages for each pair of students. Keep complete sets in order.

# Problem Pass

## Write & Solve Linear Systems In Context

### Problem 1:

Erica's Dessert Shop sells a slice of cheesecake for \$7 and a slice of pie for \$4. Yesterday she sold 205 slices of dessert for a total of \$1180.

Write and solve a system of equations to determine the number of slices of each type of dessert she sold.

# Problem Pass

## Write & Solve Linear Systems In Context

### ANSWER Problem 8:

$x$  = # bags Janet filled  
 $y$  = # bags Douglas filled

$$j + d = 16$$

$$j = 3d - 12$$

$$j = 9 \text{ bags}$$
$$d = 7 \text{ bags}$$

Janet filled 9 bags and  
Douglas filled 7 bags.

#### Original problem:

Janet and Douglas both raked leaves as a community service project with their club. Together they raked and filled 16 lawn bags of leaves. Janet filled 12 less than 3 times as many bags as Douglas.

Write and solve a system of equations to determine the number of lawn bags each person filled.

# Problem Pass

## Write & Solve Linear Systems In Context

### Problem 2:

Roy and Karen are both reading a book for their English Literature class. Roy has already read 110 pages and plans to read 45 pages a day. Karen has read 305 pages and plans to read 30 pages a day.

Write and solve a system of equations to represent this situation. After how many days will they have read the same amount of pages? What will be the page number?



# Problem Pass

## Write & Solve Linear Systems In Context

### ANSWER Problem 1:

$x$  = # slices of cheesecake  
 $y$  = # slices of pie

$$7x + 4y = 1180$$

$$x + y = 205$$

$x$  = 120 slices of cheesecake  
 $y$  = 85 slices of pie

Erica sold 120 slices of cheesecake  
and 85 slices of pie.

#### Original problem:

Erica's Dessert Shop sells a slice of cheesecake for \$7 and a slice of pie for \$4. Yesterday she sold 205 slices of dessert for a total of \$1180.

Write and solve a system of equations to determine the number of slices of each type of dessert she sold.

# Problem Pass

## Write & Solve Linear Systems In Context

### Problem 3:

Suzanne's age is two less than six times Kaitlin's age. Together their ages add up to 82.

Write and solve a system of equations to determine each person's age.

# Problem Pass

## Write & Solve Linear Systems In Context

### ANSWER Problem 2:

$x$  = # days  
 $y$  = total # pages

$$\begin{aligned}y &= 45x + 110 \\y &= 30x + 305\end{aligned}$$

OR

$$\begin{aligned}y &= 110 + 45x \\y &= 305 + 30x\end{aligned}$$

$$\begin{aligned}x &= 13 \text{ days} \\y &= \text{page } 695\end{aligned}$$

After 13 days, both  
will have read page 695.

#### Original problem:

Roy and Karen are both reading a book for their English Literature class. Roy has already read 110 pages and plans to read 45 pages a day. Karen has read 305 pages and plans to read 30 pages a day.

Write and solve a system of equations to represent this situation. After how many days will they have read the same amount of pages? What will be the page number?

# Problem Pass

## Write & Solve Linear Systems In Context

### Problem 4:

Ellen has whole milk (with 3.5% milkfat) and low fat milk (with 1% milkfat).

How many cups of whole milk and cups of low fat milk should she combine to make 20 cups of 2% milk?

Write and solve a system of equations to represent this situation.

# Problem Pass

## Write & Solve Linear Systems In Context

### ANSWER Problem 3:

$s$  = Suzanne's age  
 $k$  = Kaitlin's age

$$s = 6k - 2$$

$$s + k = 82$$

$s = 70$  years old  
 $k = 12$  years old

Suzanne is 70 years old  
and Kaitlin is 12 years old.

#### Original problem:

Suzanne's age is two less than six times Kaitlin's age. Together their ages add up to 82.

Write and solve a system of equations to determine each person's age.

# Problem Pass

## Write & Solve Linear Systems In Context

### Problem 5:

Rob bought 6 bushes and 3 trees to plant in his yard for a cost of \$768. He bought 5 more bushes and 2 more trees for \$571.

Write and solve a system of equations to determine the cost for each bush and the cost for each tree.

# Problem Pass

## Write & Solve Linear Systems In Context

### ANSWER Problem 4:

$x$  = # cups of whole (3.5%) milk  
 $y$  = # cups of low fat (1%) milk

$$0.035x + 0.01y = 0.02(20)$$

or  $0.035x + 0.01y = 0.4$

$$x + y = 20$$

$x$  = 8 cups whole (3.5%) milk  
 $y$  = 12 cups low fat (1%) milk

Ellen will need 8 cups of whole (3.5%) milk and 12 cups of low fat (1%) milk to make 20 cups of 2% milk.

#### Original problem:

Ellen has whole milk (with 3.5% milkfat) and low fat milk (with 1% milkfat).

How many cups of whole milk and cups of low fat milk should she combine to make 20 cups of 2% milk?

Write and solve a system of equations to represent this situation.

# Problem Pass

## Write & Solve Linear Systems In Context

### Problem 6:

Two angles are complementary. The measurement of one angle is 18 degrees smaller than twice the measurement of the other angle.

Write and solve a system of equations to determine the measurement of each angle.



# Problem Pass

## Write & Solve Linear Systems In Context

### ANSWER Problem 5:

$x$  = cost of a bush  
 $y$  = cost of a tree

$$6x + 3y = 768$$

$$5x + 2y = 571$$

$x$  = \$59 for a bush  
 $y$  = \$138 for a tree

Each bush cost \$59 and  
each tree cost \$138.

#### Original problem:

Rob bought 6 bushes and 3 trees to plant in his yard for a cost of \$768. He bought 5 more bushes and 2 more trees for \$571.

Write and solve a system of equations to determine the cost for each bush and the cost for each tree.

# Problem Pass

## Write & Solve Linear Systems In Context

### Problem 7:

A restaurant has large tables that seat 12 people and small tables that seat 8 people. Your team expects 260 people to attend the end-of-season banquet. If you want 28 tables in all, how many of each size table will you use?

Write and solve a system of linear equations to represent this situation.

# Problem Pass

## Write & Solve Linear Systems In Context

### ANSWER Problem 6:

$x$  = measurement of one angle  
 $y$  = measurement of the other angle

$$x + y = 90$$

$$x = 2y - 18$$

$$\begin{aligned} x &= 54 \text{ degrees} \\ y &= 36 \text{ degrees} \end{aligned}$$

One angle measures 54 degrees,  
the other measures 36 degrees.

#### Original problem:

Two angles are complementary. The measurement of one angle is 18 degrees smaller than twice the measurement of the other angle.

Write and solve a system of equations to determine the measurement of each angle.

# Problem Pass

## Write & Solve Linear Systems In Context

### Problem 8:

Janet and Douglas both raked leaves as a community service project with their club. Together they raked and filled 16 lawn bags of leaves. Janet filled 12 less than 3 times as many bags as Douglas.

Write and solve a system of equations to determine the number of lawn bags each person filled.

# Problem Pass

## Write & Solve Linear Systems In Context

### ANSWER Problem 7:

$x$  = # large tables  
 $y$  = # of small tables

$$12x + 8y = 260$$

$$x + y = 28$$

$x = 9$  large tables  
 $y = 19$  small tables

You will use 9 large tables  
and 19 small tables.

#### Original problem:

A restaurant has large tables that seat 12 people and small tables that seat 8 people. Your team expects 260 people to attend the end-of-season banquet. If you want 28 tables in all, how many of each size table will you use?

Write and solve a system of linear equations to represent this situation.

# Write & Solve Linear Systems In Context

## Problem Pass

### Answer Key for Teacher

1. Erica's Dessert Shop sells a slice of cheesecake for \$7 and a slice of pie for \$4. Yesterday she sold 205 slices of dessert for a total of \$1180. Write and solve a system of equations to determine the number of slices of each type of dessert she sold.

$$\begin{aligned} 7x + 4y &= 1180 \\ x + y &= 205 \end{aligned}$$

$$\begin{aligned} x &= 120 \text{ slices of cheesecake} \\ y &= 85 \text{ slices of pie} \end{aligned}$$

Erica sold 120 slices of cheesecake and 85 slices of pie.

2. Roy and Karen are both reading a book for their English Literature class. Roy has already read 110 pages and plans to read 45 pages a day. Karen has read 305 pages and plans to read 30 pages a day.. Write and solve a system of equations to represent this situation. After how many days will they have read the same amount of pages? What will be the page number?

$$\begin{aligned} y &= 45x + 110 \\ y &= 30x + 305 \end{aligned}$$

OR

$$\begin{aligned} y &= 110 + 45x \\ y &= 305 + 30x \end{aligned}$$

$$\begin{aligned} x &= 13 \text{ days} \\ y &= \text{page 695} \end{aligned}$$

After 13 days, both will have read page 695.

3. Suzanne's age is two less than six times Kaitlin's age. Together their ages add up to 82. Write and solve a system of equations to determine each person's age.

$$\begin{aligned} s &= 6k - 2 \\ s + k &= 82 \end{aligned}$$

$$\begin{aligned} s &= 70 \text{ years old} \\ k &= 12 \text{ years old} \end{aligned}$$

Suzanne is 70 years old and Kaitlin is 12 years old

4. Ellen has whole milk (with 3.5% milkfat) and low fat milk (with 1% milkfat). How many cups of whole milk and cups of low fat milk should she combine to make 20 cups of 2% milk? Write and solve a system of equations to represent this situation.

$$\begin{aligned} 0.035x + 0.01y &= 0.02(20) \\ \text{or } 0.035x + 0.01y &= 0.4 \end{aligned}$$

$$\begin{aligned} x &= 8 \text{ cups whole (3.5\%) milk} \\ y &= 12 \text{ cups low fat (1\%) milk} \end{aligned}$$

$$x + y = 20$$

Ellen will need 8 cups whole (3.5%) milk and 12 cups low fat (1%) milk to make 20 cups of 2% milk.

# Write & Solve Linear Systems In Context

## Problem Pass

### Answer Key for Teacher

5. Rob bought 6 bushes and 3 trees to plant in his yard for a cost of \$768. He bought 5 more bushes and 2 more trees for \$571. Write and solve a system of equations to determine the cost for each bush and the cost for each tree.

$$\begin{aligned} 6x + 3y &= 768 \\ 5x + 2y &= 571 \end{aligned}$$

$$\begin{aligned} x &= \$59 \text{ for a bush} \\ y &= \$138 \text{ for a tree} \end{aligned}$$

Each bush cost \$59 and each tree cost \$138.

6. Two angles are complementary. The measurement of one angle is 18 degrees smaller than twice the measurement of the other angle. Write and solve a system of equations to determine the measurement of each angle.

$$\begin{aligned} x + y &= 90 \\ x &= 2y - 18 \end{aligned}$$

$$\begin{aligned} x &= 54 \text{ degrees} \\ y &= 36 \text{ degrees} \end{aligned}$$

One angle measures 54 degrees and the other measures 36 degrees.

7. A restaurant has large tables that seat 12 people and small tables that seat 8 people. Your team expects 260 people to attend the end-of-season banquet. If you want 28 tables in all, how many of each size table will you use? Write and solve a system of linear equations to represent this situation.

$$\begin{aligned} 12x + 8y &= 260 \\ x + y &= 28 \end{aligned}$$

$$\begin{aligned} x &= 9 \text{ large tables} \\ y &= 19 \text{ small tables} \end{aligned}$$

You will use 9 large tables and 19 small tables.

8. Janet and Douglas both raked leaves as a community service project with their club. Together they raked and filled 16 lawn bags of leaves. Janet filled 12 less than 3 times as many bags as Douglas. Write and solve a system of equations to determine the number of lawn bags each person filled.

$$\begin{aligned} j + d &= 16 \\ j &= 3d - 12 \end{aligned}$$

$$\begin{aligned} j &= 9 \text{ bags} \\ d &= 7 \text{ bags} \end{aligned}$$

Janet filled 9 bags and Douglas filled 7 bags.

**Problem Pass Answer Sheet**  
**Write & Solve Linear Systems In Context**

Name \_\_\_\_\_

# \_\_\_\_\_ Answer \_\_\_\_\_

# \_\_\_\_\_ Answer \_\_\_\_\_

# \_\_\_\_\_ Answer \_\_\_\_\_

# \_\_\_\_\_ Answer \_\_\_\_\_

# \_\_\_\_\_ Answer \_\_\_\_\_

# \_\_\_\_\_ Answer \_\_\_\_\_



**Problem Pass Answer Sheet**  
**Write & Solve Linear Systems In Context**

Name \_\_\_\_\_

# \_\_\_\_\_ Answer \_\_\_\_\_

# \_\_\_\_\_ Answer \_\_\_\_\_

# \_\_\_\_\_ Answer \_\_\_\_\_

# \_\_\_\_\_ Answer \_\_\_\_\_

# \_\_\_\_\_ Answer \_\_\_\_\_

# \_\_\_\_\_ Answer \_\_\_\_\_